

Protechsoft day4 notes

1. Transform

The transform property is used to change the position, size, rotation, or shape of an element.

Syntax

transform: value;

translate()

Used to move an element from its original position.

Example

transform: translate(100px, 50px);

- First value → X-axis movement
- Second value → Y-axis movement

rotate()

Used to rotate an element.

Example

transform: rotate(360deg);

scale()

Used to increase or decrease the size of an element.

Example

transform: scale(0.5);

- 1 → Original size
- 0.5 → Half size
- 2 → Double size

skew()

Used to tilt an element.

Example

```
transform: skew(45deg);
```

Questions

Q1. What is transform?

Ans: Transform is used to modify the position, size, rotation, or shape of an element.

Q2. Name some transform functions.

Ans:

- translate()
- rotate()
- scale()
- skew()

2. Animation

Animation is used to create movement and visual effects on elements.

Steps to Create Animation

1. Create animation using @keyframes
2. Apply animation to an element

@keyframes

Used to define animation stages.

Example

```

@keyframes dark_light
{
    from{
        background-color: ■black;
    }
    to{
        background-color: □white;
    }
}

```

Applying Animation

```

.box{
    height: 200px;
    width: 200px;
    background-color: ■red;
    transition: all 1s ease-in;
    margin: 200px auto;
    animation: dark_light 0.3s infinite linear alternate;
}

```

Animation Properties

Property	Purpose
animation-name	Animation name
animation-duration	Animation time
animation-delay	Delay before start
animation-iteration-count	Number of repetitions
animation-direction	Direction of animation

Questions

Q1. What is animation?

Ans: Animation is used to create movement and visual effects on web pages.

Q2. Which rule is used to create animations?

Ans: @keyframes

Q3. Which property specifies animation time?

Ans: animation-duration

3. Flexbox

Flexbox is a one-dimensional layout system used to arrange elements in rows or columns.

Why Flexbox?

- Easy alignment
- Responsive layouts
- Space distribution
- Element positioning

Display Flex

```
.outer{  
  display:flex;  
}
```

Important Flex Properties

flex-direction

Determines the direction of items.

```
flex-direction: row;  
flex-direction: column;
```

justify-content

Aligns items horizontally.

```
justify-content:center;  
justify-content:space-between;  
justify-content:space-around;
```

align-items

Aligns items vertically.

```
align-items:center;
```

Questions

Q1. What is Flexbox?

Ans: A one-dimensional layout model used for arranging elements.

Q2. Which property enables Flexbox?

Ans: display:flex

Q3. Which property changes flex direction?

Ans: flex-direction

Q4. Which property aligns items horizontally?

Ans: justify-content

Q5. Which property aligns items vertically?

Ans: align-items

```
<!-- flex -->
```

```
<div class="outer">  
  <div class="card">card-1</div>  
  <div class="card">card-2</div>  
  <div class="card">card-3</div>  
</div>
```

```
.outer{  
  height: 400px;  
  background-color: red;  
  display: flex;  
  justify-content: space-around;  
  align-items: center;  
  gap: 10px;  
}
```

```
.card{  
  height: 100px;  
  width: 300px;  
  border: 2px solid black;  
  background-color: lightpink;  
}
```



4. Responsive Web Design

Responsive design makes webpages adjust automatically to different screen sizes.

Benefits

- Mobile Friendly
- Better User Experience
- Works on all devices

5. Media Query

Media queries are used to apply styles based on screen size.

Syntax

```
@media(condition){  
  
}
```

Example

```
@media(max-width:798px){  
  
  h2{  
    color:green;  
  }  
  
  .outer{  
    background-color:yellow;  
    flex-direction:column;  
  }  
}
```

Explanation

When screen width becomes **798px or less**:

- Heading color becomes green.
- Flex direction changes to column.
- Background color changes to yellow.

Questions

Q1. What is Responsive Web Design?

Ans: Designing webpages that work properly on different screen sizes.

Q2. What is a Media Query?

Ans: A CSS feature used to apply styles based on device size.

Q3. Which rule is used for responsive design?

Ans: @media

Q4. What does max-width mean?

Ans: Styles will apply when screen width is equal to or less than the specified value.